DARIO TRINCHERO

PhD Candidate in Mathematics

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EDUCATION

Doctorate in Mathematics

Jan 2023 – Nov 2026 | Stellenbosch University, ZA

Thesis: *New isomorphism between spaces of skeins & holomorphic sections in Chern–Simons theory*

Master's in Theoretical Physics

Feb 2021 – Dec 2022 | Stellenbosch University, ZA

Thesis: *Pinhole interference in 3D fuzzy space*

Courses: Quantum information, relativity, solid-state physics

Honours in Mathematics

Feb 2020 – Dec 2020 | Stellenbosch University, ZA

Mini Thesis: *Generalised Feynman formula for Ising model*

Courses: Quantum field theory, (functional) analysis

Bachelor's in Theoretical Physics

Feb 2017 – Dec 2019 | Stellenbosch University, ZA

Majors: Physics, mathematics, abstract mathematics

Extra Courses:

- *3rd year:* Algebra, logic, abstract mathematics
- *2nd year:* Applied mathematics, abstract mathematics
- *1st year:* Chemistry

Cambridge International AS- & A-Levels

Jan 2015 – Nov 2016 | Somerset College, ZA

A-Levels: Mathematics, physics, computer science, chemistry

AS-Levels: English, Afrikaans

SKILLS & INTERESTS

Some of my broad interests, ranked by my aptitude in each:

Geometry, Quantum field theory, Programming	●●●●●
Teaching, Topology, Lie theory, Category theory	●●●●○
Writing, Public speaking, Debating, Algebra	●●●◐○
Cyber-security, Graphic design, Quantum computing	●●○○○

I have been programming since 2011, and have a strong grasp of algorithms. I am comfortable with Unix and the following languages:

Python, C/C++, Java, x86 Assembly, Ruby, Octave, MATLAB, Mathematica, Web frameworks, TEX, Haskell (learning), Rust (learning)

PROFILE

I am an early-career mathematical physicist working in low-dimensional topology and Chern–Simons theory. I aspire toward a position that combines research with lecturing and supervision.

I am passionate about teaching: I have tutored undergraduate mathematics since 2018 and lectured a large first-year class in 2024, whose students nominated me for a ROCTAS Award as their most impactful lecturer. I see teaching as a means to give back the guidance and opportunities I was privileged to receive in my own education.

Hobbies

I enjoy reading, cooking, trail-running, rock-climbing, and tabletop games, especially the abstract strategy game Hive.

Citizenship

I am a dual citizen of South Africa and Italy.

KEY ACCOLADES

- Globally-competitive results in university courses, including Dean's Medal
- Numerous awards from Stellenbosch University, including for teaching
- Graduated all degrees *cum laude*, including being top of my undergraduate class
- Author of academic research published in the Annals of Physics

REFEREES

Dr Bruce Bartlett (Hons & PhD supervisor)
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Mathematics department
Stellenbosch University, South Africa

Prof. Frederik Scholtz (MSc supervisor)
fgs@sun.ac.za
Physics department
Stellenbosch University, South Africa

Dr Johannes Kriel (MSc examiner)
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Physics department
Stellenbosch University, South Africa

RESEARCH

Journal Articles

- **D. Trinchero** and F. G. Scholtz, "Pinhole interference in three-dimensional fuzzy space," *Annals of Physics*, vol. 450, p. 169 224, Mar. 2023. DOI: 10.1016/j.aop.2023.169224

Theses

- **D. Trinchero**, "Pinhole interference in three-dimensional fuzzy space," MSc thesis, Stellenbosch University, Mar. 2023. HDL: 10019.1/126944.

Talks & Posters

- **D. Trinchero**, *A new isomorphism between the spaces of skeins & holomorphic sections in Chern–Simons theory*, Poster, CMND, University of Notre Dame, Jun. 2024.
- **D. Trinchero**, *Introduction to quantum groups*, Stellenbosch University, Oct. 2023. DOI: 10.5281/zenodo.10000346
- **D. Trinchero**, *Tour of knots & theta functions*, Stellenbosch University, Oct. 2023. DOI: 10.5281/zenodo.10047936
- **D. Trinchero**, *Pinhole interference in 3D fuzzy space*, Syracuse University, Sep. 2023. [Online]. Available: https://video.syr.edu/media/t/1_74dkljgj
- **D. Trinchero**, *Computing by collapsing*, Cosmic Conversations, Stellenbosch University, Apr. 2022. DOI: 10.5281/zenodo.8228648

Complete list: https://dariotrinchero.github.io/talks/

ACHIEVEMENTS

2023–2026: Doctorate in Mathematics

- ROCTAS Award: students nominate their most impactful lecturer (2025)
- Bursary: National Research Foundation

2021–2022: Master's in Theoretical Physics

- 100% for 1 of 3 modules; median mark of 97%; 89% for thesis
- Bursaries: Harry Crossley Foundation & Skye Foundation

2020: Honours in Mathematics

- 100% for 1 of 11 modules; median mark of 94%; 95% for mini thesis
- Dean's Medal: highest faculty average across 4 years of study
- Perimeter Institute for Theoretical Physics: attended summer program
- Bursary: National Research Foundation

2017–2019: Bachelor's in Theoretical Physics

- 100% for 16 of 37 modules; median mark of 99%
- Rector's Award: among top students of faculty
- Rubbi Book Prize: top Mathematics student (2018, 2019)
- Top Computer Science & Applied Mathematics first year
- First Year Achievement Awards: among top first year students
- Winning team: South African Mathematical Modelling Contest (2018)
- Second place team: *SANReN* Cyber Security Challenge (2018)
- Bursary: Stellenbosch University Merit Award (2018, 2019)

2015–2016: Cambridge A-levels

- Dux scholar: Somerset College (Cambridge curriculum)
- Top marks for AS-levels: Mathematics (globally), Chemistry (South Africa)

TEACHING

Junior Mathematics Lecturer

Feb 2024 – Jul 2024 — Stellenbosch University, ZA

I lectured calculus part-time to a class of around 220, and set tutorial & exam questions. My students nominated me for a ROCTAS Award in 2025.

Mathematics Course Assistant

Feb 2018 – present — Stellenbosch University, ZA

I tutored the following undergraduate classes:

- *3rd year:* Algebra, real analysis, Fredholm theory
- *2nd year:* Linear algebra, advanced calculus, analysis
- *1st year:* Linear algebra, calculus

Private Tutoring

Aug 2017 – Mar 2018 — Independent, *ad hoc*

Grade 11–12 mathematics and physics, both national curriculum & advanced programme mathematics.

EMPLOYMENT

AI Safety Research Fellowship

Nov 2026 – Feb 2027 — PIBBSS, Cape Town, ZA

I was selected for the PIBBSS AI safety fellowship to research computational learning theory, mentored by Vanessa Kosoy.

Software Development Engineer Intern

Nov 2019 – Feb 2020
Dec 2020 – Feb 2021 — Amazon Web Services, ZA

I interned twice for *Amazon Web Services* (AWS) where I worked on *Elastic Compute Cloud* (EC2).

PERSONAL PROJECTS

Learning isiXhosa (since 2025)

- I am self-studying isiXhosa, a Bantu language of South Africa, widely spoken in my home city of Cape Town.

Exploring Philosophy (since 2021)

- I have an avid interest in philosophy, and spend a lot of free time working through papers & textbooks.
- I presented a talk in 2024 at the Stellenbosch philosophy club on a famous (alleged) mathematical proof of God.

Game Development (since 2019)

- I partake in the annual *Global Game Jam*: teams design & build a computer game from scratch in 3–5 days.
- I have since independently created several game prototypes, including a web version of Hive (see my GitHub).

Coding Challenges (since 2015)

- An ongoing project of mine is completing programming challenges from the *Project Euler* database.
- These challenges demand mathematical proficiency & creativity to solve in the permitted 1min run-time.